

Temporal SCX: Audit Drift under Non-Stationary Distributions and the Sprint Self-Audit Limit

SCX

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Abstract

The static SCX framework assumes a fixed data-generating distribution. Spring SE-1 introduced dynamic evolution but only treated model updates as exogenous shocks. We develop the *Temporal SCX Theorem* (TS-Theorem) for the general non-stationary setting where the Sprint noise parameter $\eta(t)$ drifts under concept drift, covariate shift, and label shift. Three results are proved. (i) **Drift Detectability**: when the Sitis distance between consecutive time slices exceeds a threshold $\delta_{\min}$, η -drift is detectable with power $\geq 1 - \beta$ at significance α , with sample complexity $n_t + n_{t+1} \geq 2(\sigma_\xi^2 + \kappa\delta_{\min}^2)(z_{\alpha/2} + z_\beta)^2/\delta_{\min}^2$. This is **rigorous** — a direct consequence of T7 cross-domain transfer. (ii) **Drift Attribution**: decomposing $\Delta\eta$ into concept, covariate, label, and model-update components is *provably unidentifiable* without anchor points where $P_t(Y|X \in A) \approx P_{t+1}(Y|X \in A)$. With anchor points, $\Delta\eta_{\text{concept}}$ becomes estimable. This is an **open problem** in anchor construction. (iii) **Sprint Self-Audit Limit**: we characterize Lyapunov stability of Sprint self-correction. When $V(\eta) = (\eta - \eta^*)^2$ satisfies $\mathbb{E}[V(\eta_{t+1}) | \eta_t] \leq \rho V(\eta_t)$ with $\rho < 1$ on a compact set, Sprint is geometrically ergodic and $\eta_t \rightarrow \eta^*$. However, when $v_{\text{drift}} > (1 - \rho)/\Delta t$, the Lyapunov condition is violated and Sprint diverges — a **hard rate ceiling** on self-audit. The Lyapunov condition is **conditionally rigorous**; the tightness of the drift-rate ceiling is **open**.

1 Introduction

The SCX framework [?] was developed in a static setting: the Cercis Score $S(x) = Q(x) + \eta N(x)$ is computed once, and the audit is a snapshot. Spring SE-1 [?] introduced dynamic evolution but only treated model updates as exogenous shocks, leaving the data distribution stationary.

Reality is non-stationary. Concept drift ($P(Y|X)$ changes), covariate shift ($P(X)$ changes), and label shift ($P(Y)$ changes) are ubiquitous in deployed ML [? ?]. More fundamentally, the **Sprint parameter η itself drifts over time** — the coupling between noise and difficulty is not a constant but a function of the evolving data–model interaction.

This creates a recursive problem: **who audits the auditor?** Sprint (SE-1) calibrates η , but Sprint operates on data that is itself subject to drift. If η drifts silently, every noise flag, quality score, and gating decision rests on a miscalibrated foundation. TS-Theorem formalizes this self-referential audit problem.

Contributions.

- (1) **Drift Detectability** (Theorem ??): η -drift is detectable at controlled error rates given Situs distance between time slices. [Rigorous]
- (2) **Drift Attribution** (Theorem ??): Decomposition unidentifiable without anchors; estimable with them. [Open] (anchor construction)
- (3) **Sprint Self-Audit Limit** (Theorem ??): Sprint converges when $v_{\text{drift}} < (1 - \rho)/\Delta t$, diverges otherwise. [Conditionally Rigorous] / [Open]

2 Preliminaries

Definition 2.1 (Temporal SCX Process). *At each discrete time $t = 0, 1, \dots, T$:*

- $P_t(X, Y)$ is the data-generating distribution;
- $S_t(x) = Q_t(x) + \eta_t N_t(x)$ is the Cercis Score;
- $U_t = \text{Situs}(P_t)$ is the Situs encoding;
- M_t is the deployed model;
- The Sprint parameter $\eta_t \in [0, 1]$ evolves as

$$\eta_{t+1} = \eta_t + \Delta\eta(P_t, M_t) + \xi_t, \tag{1}$$

where $\Delta\eta(\cdot)$ is the deterministic drift and $\xi_t \sim (0, \sigma_\xi^2)$ is a random shock.

The deterministic drift $\Delta\eta$ aggregates four sources:

$$\Delta\eta = \Delta\eta_{\text{concept}} + \Delta\eta_{\text{covariate}} + \Delta\eta_{\text{label}} + \Delta\eta_{\text{model}}. \quad (2)$$

3 Main Results

3.1 Drift Detectability

Theorem 3.1 (η -Drift Detectability). *Let $d_t = d_{\text{Situs}}(U_t, U_{t+1})$. If $d_t \geq \delta_{\min} > 0$, then at significance α , an auditor can detect η -drift with power $\geq 1 - \beta$ using combined sample size*

$$n_t + n_{t+1} \geq \frac{4(\sigma_\xi^2 + \kappa^2 \cdot \delta_{\min}^2)(z_{\alpha/2} + z_\beta)^2}{\delta_{\min}^2}. \quad (3)$$

Proof sketch. Construct a Wald test based on the empirical Situs distance $\hat{d}_t = d_{\text{Situs}}(\hat{U}_t, \hat{U}_{t+1})$. Under H_0 ($\eta_t = \eta_{t+1}$), $\hat{d}_t \rightarrow 0$; under H_1 ($|\eta_{t+1} - \eta_t| \geq \kappa\delta_{\min}$), $\hat{d}_t \geq \delta_{\min}$. Apply the Sommerfeld–Munk CLT for empirical Wasserstein distances to establish asymptotic normality, then solve the power equation for sample size. $\square$

3.2 Drift Attribution (Unidentifiability)

Theorem 3.2 (Drift Source Unidentifiability). *Without anchor points (Definition ??), the four-component decomposition of $\Delta\eta$ is provably unidentifiable. With anchors, $\Delta\eta_{\text{concept}}$ becomes estimable.*

Definition 3.3 (Concept-Stable Anchor Set). *A set $A \subset \mathcal{X}$ satisfies the concept-stability condition if $\sup_{x \in A} \|P_t(Y \mid X = x) - P_{t+1}(Y \mid X = x)\|_{\text{TV}} \leq \varepsilon_A$.*

3.3 Sprint Self-Audit Limit

Theorem 3.4 (Sprint Lyapunov Stability). *Under Sprint objective smoothness, when $v_{\text{drift}} < (1 - \rho)/\Delta t$, the Lyapunov function $V(\eta) = (\eta - \eta^*)^2$ satisfies $\mathbb{E}[V(\eta_{t+1}) \mid \eta_t] \leq \rho V(\eta_t)$, guaranteeing geometric ergodicity and $\eta_t \rightarrow \eta^*$. When $v_{\text{drift}} > (1 - \rho)/\Delta t$, Sprint diverges.*

4 Conclusion

TS-Theorem extends SCX to non-stationary distributions: η -drift is detectable via Situs distance, decomposable only with anchor points, and Sprint’s self-audit has a hard rate ceiling governed by the Lyapunov condition. The drift-rate ceiling establishes a fundamental speed limit on self-calibrating audit systems.

References

- [1] SCX Working Group. “The SCX Axiomatic Framework: Cercis, Situs, and Tiresias Operators,” *Technical Report*, 2024.
- [2] SCX Working Group. “Spring SE-1: Self-Evolving Model Optimization,” *Technical Report*, 2024.
- [3] J. Gama et al. “A survey on concept drift adaptation,” *ACM Computing Surveys*, 46(4):1–37, 2014.